



COMSATS University
Islamabad
(Lahore Campus)

CSC356- Human Computer Interaction

PACT ANALYSIS IN HUMAN COMPUTER INTERACTION

Learning Objectives

1

- Understand why PACT analysis is important in HCI.

2

- Learn how to conduct a PACT analysis.

3

- Explore an example of PACT analysis.

4

- Evaluate the advantages and shortcomings of PACT analysis.

Introduction

People use **technologies** to undertake **activities** in different **contexts**.

Examples:

- Teenagers use mobile phones to send text messages while sitting on a bus.
- Secretaries use Microsoft Word to write documents in offices.

These examples show that all four elements (People, Activities, Contexts, Technologies) are always linked. If any one changes, the entire interaction changes..

PACT analysis helps designers look at all four elements(People, Activities, Contexts, and Technologies)together, so that systems remain usable and effective.

What is PACT Analysis?

PACT analysis is a **framework** for **understanding** and **designing interactions** between **people** and **technology**.

It helps you **consider all the key factors** that **influence how a system is used and experienced**.

The PACT analysis is based on the idea that **technology** should be designed to support **people activities** in the **context** in which they are being performed.

How to Conduct PACT Analysis

Step 1: Analyze People

- Identify who the users are, their abilities, and what they expect.

Step 2: Identify Activities

- Define the tasks, their difficulty, and how often they are done.

Step 3: Examine Contexts

- Note where, when, and under what conditions the system is used.

Step 4: Evaluate Technology

- Select suitable devices and consider technical limits.

PACT Analysis

PACT stands for:

People – Who are the users?

Activities – What tasks will users perform?

Contexts – Where and how will users interact with the system?

Technology – What tools, platforms, and devices are involved?



People : Who Are We Designing For?

This section is all about **understanding the users** of the system -Their **characteristics, skills, and needs**.

Key Questions to Ask:

Who are the users? (e.g., age, gender, background)

What are their skills and experience levels? (e.g., experienced, beginner)

What are their physical and cognitive abilities? (e.g., vision, attention span, cognitive load)

What are their motivations and attitudes? (e.g., do they want to use this system?)

People : Who Are We Designing For?

Example: Designing a Movie Ticket App

Users: People of all ages, from teenagers to seniors.

Skills: Varies widely; some are experts with apps, others may struggle.

Physical Abilities: Some may have difficulty reading small text, pressing small buttons, or remembering multiple steps.

Motivations: They want a quick, easy way to book a movie ticket without standing in line.

Activities: What Are Users Doing?

This part focuses on the **tasks that users need to perform** using the system. It's about what they do, not just how they do it.

Key Questions to Ask:

What are the main tasks users perform?

How often are these tasks done?

Are the tasks simple or complex?

Do the tasks need to be completed quickly?

Activities: What Are Users Doing?

Example: Designing a Movie Ticket App

1: What are the main tasks users perform?

- Searching for a movie.

- Choosing a showtime and cinema.

- Selecting seats on a seating plan.

- Paying for the tickets.

- Receiving a digital ticket.

2: How often are these tasks done?

- Frequent users book weekly or more.

- Occasional users book only every few months.

- Peak times are weekends, holidays, and big releases.

Activities: What Are Users Doing?

3. Are the tasks simple or complex?

Simple tasks: Browsing movies, checking show times, watching trailers.

Moderately complex tasks: Seat selection and secure online payments (requires attention, but users expect quick flow).

More complex (less frequent): Ticket cancellations, refund tracking, applying vouchers.

4: Do the tasks need to be completed quickly?

Yes, speed matters.

- Seats must be booked in seconds before they sell out.
- Payment/confirmation should be instant.

Context: Where Are They Using It?

Context is the **environment** in which the interaction takes place. This includes **physical, social, and organizational factors**.

Key Questions to Ask:

Physical environment: Where is the system used? (e.g., office, home, a noisy cafe..)

Social: Are users alone or with others?

Organizational: Are there rules or procedures that affect how the system is used?

Context: Where Are They Using It?

Example: Designing a Movie Ticket App

Physical: Users might be using the app on a moving bus, at home on their couch, or even while walking.

Social: A user might be booking tickets for a group of friends, so sharing information or coordinating is important.

Organizational: The app must follow the cinema's booking rules and policies.

Technologies: What Are the Tools?

This section is about the **tools and technologies** used to perform the activities. It includes both hardware and software.

Key Questions to Ask:

What hardware is being used? (e.g., smartphone, desktop computer)

What software is involved? (e.g., the app itself, payment gateways)

What other technologies are part of the ecosystem? (e.g., Wi-Fi, mobile data)

What are the limitations and capabilities of these technologies?

Technologies: What Are the Tools?

Example: Designing a Movie Ticket App

Hardware: A mobile phone (iOS and Android).

Software: The movie ticket app, a secure payment system (like a credit card processor)..

Other Technologies: The app needs a reliable internet connection (Wi-Fi or cellular data) to function.

Limitations: Screen size can be small, and connectivity might be unstable.

Conclusion of PACT Analysis – Movie Ticket App

The PACT analysis shows that designing a movie ticket app - requires balancing **different users, quick booking tasks, varied usage contexts, and the limitations of mobile technology.**

People: Wide age range → needs accessible design (large fonts, clear navigation, simple flows).

Activities: Key tasks (search, select, pay, retrieve) → must be quick and seamless.

Context: Used in varied places (home, work, bus) → The app must allow one handed use, handle different lighting conditions, and support group bookings and sharing.

Technology: Small screens & weak internet → needs zoom support and resume smoothly if the connection drops.

Why do we need PACT Analysis?

01

Ensure the design meets user needs.

02

Identify usability issues early.

03

Improve accessibility and inclusivity.

04

Optimize user experience (UX).

05

Guide the selection of appropriate technology for a given context.

Advantages

- **Encourages user-centered design.**
- Identifies usability issues early.
- Supports inclusive and accessible design.
- Adapts to different environments.
- Improves overall user experience.

Shortcomings

- Can be time consuming.
- Requires extensive user research.
- May not fully predict unexpected user behaviors.
- Limited in addressing rapid technological advancements.

Case Study: Designing a Mobile Banking Application

A banking institution plans to launch a mobile banking app for customers. The purpose of this app is to allow users to manage their bank accounts and perform financial transactions directly from their smartphones, without visiting a physical branch.

Questions for PACT Analysis:

People – Who are the customers that will use this app? What are their characteristics, skills, and goals?

Activities – What kinds of banking tasks will customers perform using the app? How often will these tasks be done, and how complex might they be?

Context – In what situations and environments will customers typically use this app?

Technology – What devices and technologies will support this app? What technical limitations or strengths should be considered?

Case Study: Designing a Mobile Banking Application

People

Users: Bank customers (18–70+, novice to expert)

Needs & goals: Quick, safe access to banking (checking balance, paying bills, transferring money).

Motivation: Convenience (avoid branch visits).

Accessibility: Users with low vision or physical limitations may need larger text, voice support, or simplified flows.

Activities

Simple: Balance check, viewing transactions

Moderate: Fund transfers, bill payments

Complex: Loan applications, card management

Frequency: Daily (balance check), monthly (bills), occasional (loans)

Case Study: Designing a Mobile Banking Application

Context: Usable in different environments (home, travel, work) and allow safe group usage.

Technology

Hardware: Smartphones.

Software: (Mobile banking app itself, Integration with payment gateways (Visa, MasterCard, local wallets like Easy paisa/Jazz Cash), Biometric authentication (fingerprint)).

Case Study: Designing a Mobile Banking Application

Other technologies: Internet connectivity...

Limitations & challenges:

Small screen size: Users may misclick while entering amounts.

Solution: large, clear buttons and confirmation prompts.

Unstable internet: Transaction may fail midway .

solution: auto-resume and “transaction in progress” notifications.

References

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